

Andrei-Cătălin Popa

PHD STUDENT IN ASTROPHYSICS

Lennard-Jones School of Chemical and Physical Sciences, Keele University, Staffordshire, ST5 5BG, UK

✉ andrei-catalin.popa@students.uniroma2.eu | 🌐 acpopa.github.io | 📧 acpopa

Education

Keele University

PHD IN ASTROPHYSICS

Supervisor: Dr. Pierre Maxted.

Thesis: *Calibrating Stellar Models with Chemical Tracers of Internal Mixing Processes.*

PLATO (launch 2026) aims to discover and characterise rocky planets orbiting Sun-like stars, with stellar ages measured to 10% accuracy. Developing a new method to calibrate next-generation stellar models using chemical tracers in binary star spectra to probe the efficiency of internal mixing processes (convective overshooting, atomic diffusion, and rotation) in Sun-like stars of known mass.

Keele, UK

Oct. 2026 – present

Université Côte d'Azur

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Thesis: *Characterising M-dwarf Stars as Exoplanet Hosts in Preparation for the PLATO Mission.*

Advisors: Dr. Orlagh Creevey (OCA), Dr. Valentina D'Orazi (Tor Vergata), Dr. Ulrike Heiter (Uppsala).

Research Internship: *Fundamental Parameters of FGK Stars.*

Advisors: Dr. Orlagh Creevey (OCA) and Dr. Caroline Soubiran (Bordeaux).

Nice, France

Mar. 2025 – Sept. 2026

Università degli Studi di Roma "Tor Vergata"

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME) — FIRST SEMESTER

Consortium of four universities: Università degli Studi di Roma "Tor Vergata", Universität Bremen, Université Côte d'Azur, and University of Belgrade.

Rome, Italy

Sept. 2024 – Mar. 2025

University College London

MSCI IN PHYSICS (FIRST CLASS HONOURS)

Supervisor: Prof. Daisuke Kawata (Mullard Space Science Laboratory).

Thesis: *Hunting for Intermediate-Mass Black Hole Binaries.*

London, United Kingdom

Sept. 2018 – Jun. 2022

Research Experience

Astrophysics Research Centre

PHD RESEARCH: CALIBRATING STELLAR MODELS WITH CHEMICAL TRACERS OF INTERNAL MIXING PROCESSES

- Supervised by Dr. Pierre Maxted. Research within the ESA PLATO mission consortium, focused on using chemical abundance tracers in binary stars to calibrate stellar structure and evolution models.
- Investigating internal mixing processes (convective overshooting, atomic diffusion, rotation) through precision constraints provided by binary star systems and PLATO lightcurves.

Keele, United Kingdom

Oct. 2026 – present

Observatoire de la Côte d'Azur

MASTER'S THESIS: CHARACTERISING M-DWARF STARS AS EXOPLANET HOSTS – PREPARATION FOR THE PLATO

MISSION

- Supervised by Dr. Orlagh Creevey (OCA), Dr. Ulrike Heiter (Uppsala University) and Dr. Valentina D'Orazi (Tor Vergata), within the Erasmus Mundus Joint Master's in Astrophysics and Space Science (MASS).
- Deriving luminosities and radii of benchmark M-dwarf stars via spectral energy distribution (SED) fitting and bolometric flux calculations, using Gaia XP spectra and multi-band photometry.
- Building and testing Python pipelines for photometric and spectroscopic data comparison, integrating literature stellar parameters for cross-validation.
- Contributing to PLATO host-star modelling by refining fundamental parameters used in the mission's stellar characterisation framework.

Nice, France

Jan. 2026 – Sept. 2026

Astrophysics Research Institute

SUMMER RESEARCH PROJECT: WHITE DWARF COOLING SEQUENCES AND INFRARED COLOUR EXCESS

- Supervised by Prof. Maurizio Salaris. Investigated whether the apparent broadening of the white dwarf cooling sequence in NGC 6397 reflects an intrinsic IR-excess population or statistical scatter.
- Analysed JWST and HST photometric data; constructed synthetic white dwarf populations from theoretical isochrones and a Salpeter IMF, with Monte Carlo uncertainty propagation.
- Developed Python routines to compare observed and simulated colour-magnitude diagrams, applying Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) analysis.

Liverpool, United Kingdom

Jul. 2025 – Sept. 2025

Scholarships & Honours

2026–2030	STFC PhD Studentship , Science and Technology Facilities Council / UKRI	£21,805 p.a.
2024–2026	Erasmus Mundus (EMJM) Scholarship , European Education and Culture Executive Agency	€33,600
2022	Summer Research Internship Scholarship , Dept. of Physics & Astronomy, UCL	£2,000
2022	"Teach Physics" Internship Award , Ogden Trust — Eastbury Community School, London	£1,400

Teaching

Eastbury Community School

Barking, United Kingdom

SUMMER PHYSICS TEACHING INTERN — OGDEN TRUST "TEACH PHYSICS" PROGRAMME

Jun. – Jul. 2022

Taught A-level and GCSE physics through interactive sessions and lab exercises; created classroom materials aligned with UK physics curricula; supported Science Club activities.

Outreach

UCL Observatory

Mill Hill, United Kingdom

DEMONSTRATOR

Oct. 2019 – Sept. 2022

Trained to operate robotic and manual telescopes (14-inch Schmidt–Cassegrain; twin 24/18-inch refractors). Led public observing sessions for school groups and members of the public; developed science communication skills across diverse audiences.

Department of Physics & Astronomy, University College London

London, United Kingdom

OUTREACH INTERN

Jun. – Sept. 2022

Organised outreach activities in local schools and community groups; planned and delivered science communication events including UCL's public festival *Your Universe*; produced materials translating astrophysical research for non-specialist audiences.

Service & Committees

2024–2025 **Student Representative**, Quality Evaluation Committee, Erasmus Mundus MASS Programme

Rome/Nice

Skills & Professional Memberships

Programming	Python, MATLAB, IDL
Astronomical Software	TOPCAT, SAO DS9, AstroImageJ, VOSA, iSpec
Tools & Environments	Git, $\LaTeX$, UNIX/Linux
Professional Memberships	Institute of Physics (IOP); Mensa Romania; England Handball Referee Association
Languages	English: Fluent Romanian: Native French: Intermediate

References

Dr Pierre Maxted

Keele University

READER IN ASTROPHYSICS

contact: p.maxted@keele.ac.uk

Dr Orlagh Creevey

Observatoire de la Côte d'Azur

RESEARCH ASTRONOMER

contact: orlagh.creevey@oca.eu

Dr Valentina D'Orazi

Università di Roma "Tor Vergata"

TENURE-TRACK RESEARCHER IN ASTROPHYSICS

contact: vdorazi@roma2.infn.it